

Module 0 Checkpoint

Getting Started with Logging onto CS Servers

What to submit?

Please submit a document in the form of .doc/.docx/.pdf to Moodle, with the required TWO screen captures for Task 2.

HKUVPN

The following tasks require your computer to be connected to the HKUVPN. Procedures for making HKUVPN connection can be found at <https://its.hku.hk/kb/user-guide-on-making-hkuvpn-connection-with-mfa/>.

For any technical issue about connecting to HKUVPN, you should contact HKU ITS (<https://www.its.hku.hk/service-desk>). After the HKUVPN connection is made, you can proceed to the following tasks.

Task 1:

In order to log in to computers in our (a) student labs (b) the academy servers, you are required to obtain a CS computer account.

Before you start to get one, connect your device to **HKUVPN** to gain access to the CS Intranet.

1. Visit CS homepage (<http://www.cs.hku.hk>).
2. Click “Intranet”.
3. Click “New Student”.
4. Log in using your HKU Portal account.
5. After approval, you should be able to see your CS username, password and a door PIN (for accessing nonpublic CS labs) on the page.

In the document that you will submit, please include your CS username.

Task 2:

Note: you would not be able to finish this task if you could not obtain your CS computer account from **Task 1**.

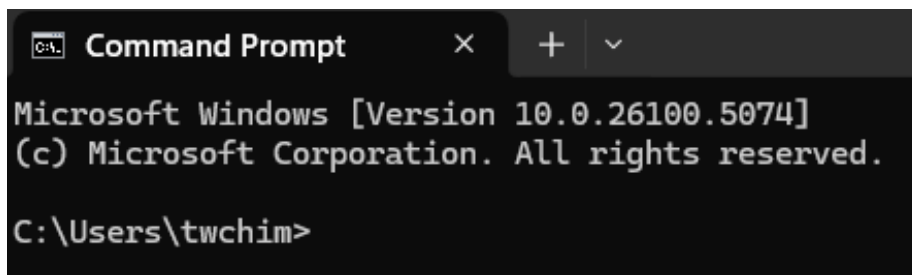
For learning purposes, you should remote access the Ubuntu Linux server (i.e., the academy server) in CS department via SSH (without GUI) and SFTP. Please follow the procedures below to connect to the Linux server in the CS department.

SSH

SSH provides you with a command-line user-interface to interact with the remote computer. The academy servers in our CS department can be accessed by using an SSH connection. The hostnames are academy11.cs.hku.hk and academy21.cs.hku.hk. However, they do not allow direct access from a computer outside the CS Network. If you are connecting the server from outside the CS network (e.g., from home), you should first make an **HKUVPN** connection. If you are connecting the server from campus network (e.g., via HKU WiFi), you should first connect to gatekeeper.cs.hku.hk before connecting to academy11.cs.hku.hk or academy21.cs.hku.hk.

Most of the operating systems come with SSH clients. You may use Command Prompt or Windows PowerShell in Windows 10/11 or Terminal in macOS. The following procedures apply to the SSH clients mentioned above. If you are using Windows 10, update the OS to the latest version to obtain the SSH feature. For older versions of Windows, third-party SSH client such as PuTTY (<https://www.putty.org/>) can be used.

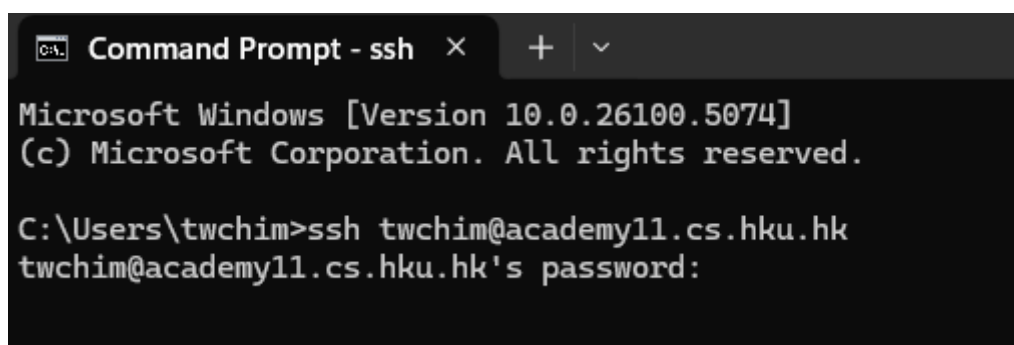
1. Open an SSH Client (Command Prompt or Windows PowerShell or Terminal).



```
Microsoft Windows [Version 10.0.26100.5074]
(c) Microsoft Corporation. All rights reserved.

C:\Users\twchim>
```

2. Enter “ssh *cs_account*@academy11.cs.hku.hk” (where *cs_account* is your CS computer account) and then enter the password. When typing your password, it is normal that your password is not being shown. Press Enter when finished typing.



```
Microsoft Windows [Version 10.0.26100.5074]
(c) Microsoft Corporation. All rights reserved.

C:\Users\twchim>ssh twchim@academy11.cs.hku.hk
twchim@academy11.cs.hku.hk's password:
```

Password will NOT be shown when typing

If you are connecting the server from campus network (e.g., via HKU WiFi), you should first enter “ssh *cs_account*@gatekeeper.cs.hku.hk” (where *cs_account* is your CS computer account) and then enter the password. After that, proceed to enter “ssh *cs_account*@academy11.cs.hku.hk” (where *cs_account* is your CS computer account) and then enter the password. That is, 2 levels of login are required.

3. After login, it might ask you “Terminal type?”. Just press Enter to bypass the message. Then, you should see you are logged in to the academy server (@academy11 is shown in the example below). Please ignore all upgrade messages as only the system administrator can launch any system upgrade. Bingo, you already entered our departmental server!!!

```
C:\Users\twchim>ssh twchim@academy11.cs.hku.hk
twchim@academy11.cs.hku.hk's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-60-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep  3 16:15:54 HKT 2025

System load:   0.46      Temperature:   65.0 C
Usage of /home: unknown  Processes:    2798
Memory usage:  3%       Users logged in: 1
Swap usage:    0%       IPv4 address for eth0: 147.8.178.21

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Get cloud support with Ubuntu Advantage Cloud Guest:
http://www.ubuntu.com/business/services/cloud

Expanded Security Maintenance for Applications is not enabled.

11 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

21 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon Aug 18 17:10:21 2025 from 147.8.179.11
twchim@academy11:~$ |
```

4. Done! **Try entering the first Linux command date, what can you see? Take a screenshot of your command-line window and attach it to the document.**

SFTP (Recommended for file transfers)

SFTP allows you to transfer files between your local computer and the remote server. The hostname of the academy server is **sftp://academy.cs.hku.hk**. Similar to SSH, it does not allow direct access from a computer outside the CS Network. If you are connecting the server from outside the CS network (e.g., from home), you should first make an **HKUVPN** connection.

Please try using a SFTP client to access the academy server and take a screenshot of your home directory and attach it to the document. If you have no idea what a SFTP client is, please follow the procedures below to use FileZilla to access the server.

Using FileZilla to access to the academy server

The following procedures are based on using FileZilla but you are welcome to use other SFTP clients.

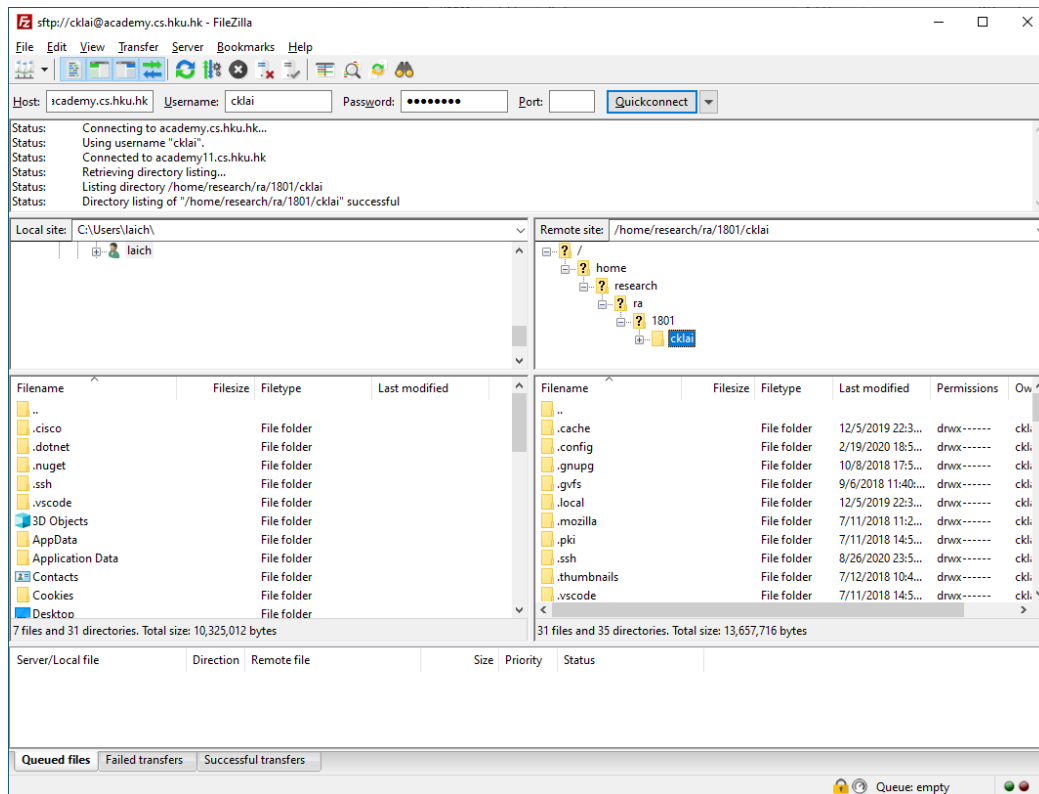
1. Download and install FileZilla Client

- <https://filezilla-project.org/>
- Please follow its procedures for installing **FileZilla Client**.
- **Be careful:** The installer may include options for installing other sponsored software. Uncheck or decline them if you don't want to install those software.

2. Open FileZilla, fill in the following information and click "Quickconnect"

- Host: sftp://academy.cs.hku.hk
- Username: [your_CS_account]
- Password: [your_CS_account_password]
- Port: Leave it empty

3. After connection, the window on the right side shows your home directory on the academy server. You can now drag and drop files between your local computer and the server. If the connection fails, check if you have connected to HKUVPN.



Do not forget to take a **screenshot** of your home directory and attach it to the document.

This checkpoint serves to make sure that you can get access to the CS servers, which would be the major environment that you will be working in for this course. You will learn more about the Linux environment in the next two modules.

